Time Identification of Non-Commensurable Fractional Order Systems of the Second Kind

This is the Thesis Subtitle if necessary.

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Thesis to obtain the Master of Science Degree in

Mechanical Engineering

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

I would like to thank my parents for their friendship, encouragement and caring over all these years, for always being there for me through thick and thin and without whom this project would not be possible. I would also like to thank my grandparents, aunts, uncles and cousins for their understanding and support throughout all these years.

To each and every one of you – Thank you.

Abstract

lorem ipsum

Keywords

Maecenas tempus dictum libero; Donec non tortor in arcu mollis feugiat;Cras rutrum pulvinar tellus.

Resumo

lorem ipsum

Palavras Chave

Colaborativo; Codificaçãoo; Conteúdo Multimédia; Comunicação;

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Listings

Acronyms

FOTF Fractional-Order Transfer Function

HD High Definition

SD Standard Definition

SS Steady-state

WLAN Wireless Local Area Network

WWAN Wireless Wide Area Network

1

Introduction

Todo list

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State Of The Art

3

Mathematical Modeling of Fractional Order Systems

Contents

Many real-world phenomena, such as anomalous diffusion and viscoelasticity cannot be easily modeled by integer-order equations. Fractional Calculus tackles this problem by expanding the concepts of differentiation and integration to non-integer orders that better represent the systems mentioned before.

This mathematical extension leads to the definition of the Fractional-Order Transfer Function (FOTF), where integer powers of the Laplace variable *s* are replaced by real-valued exponents.

3.1 General Fractional-Order Transfer Functions

A general fractional-order system can be represented by a transfer function like the following:

$$G(s) = \frac{b_m s^{\beta_m} + b_{m-1} s^{\beta_{m-1}} + \cdots + b_0 s^{\beta_0}}{a_n s^{\alpha_n} + a_{n-1} s^{\alpha_{n-1}} + \cdots + a_0 s^{\alpha_0}} \tag{3.1}$$

where α_k, β_k are non-negative real numbers. These exponents provide an additional degree of freedom for modeling complex dynamics.

3.1.1 Commensurate vs. Non-Commensurate Systems

Fractional Order systems are categorized by the relationship between their differentiation orders, *i.e.*, the values of the transfer function's exponents:

1. **Commensurate Order Systems:** A system is said to be commensurate order if all the exponents ν_k are integer multiples of a single base order ν.

2. **Non-Commensurate Order Systems:** A system is non-commensurate when its exponents cannot be expressed as multiples of a common base order.

add references

3.2 The Second Species Model

This research will be focused on a specific class of fractional-order models known as "second species" systems , since they resemble second-order transfer functions. In this case, the differential orders are ν and $\nu + 1$ as shown below :

Reference malti

reference hmed since he studied this in detail

$$G(s) = \frac{K e^{-\theta s}}{1 + 2\zeta \left(\frac{s}{\omega_n}\right)^{\nu} + \left(\frac{s}{\omega_n}\right)^{\nu+1}} \tag{3.2}$$

where the parameters are defined as follows:

- K: Gain of the system;

- $e^{-\theta s}$: delay time;
- $\nu \in \mathbb{R}^+$: the fractional order of the system;
- ζ: the pseudo-damping coefficient;
- ω_n: the natural frequency of the system.

3.2.1 Structural Analysis

This model is classified as non-commensurate since the powers of the Laplace variable s (in this case, ν, $\nu + 1$) are not necessarily multiples of a single base order unless ν itself is an integer, which defeats the purpose. This structural choice is important because it ensures the natural frequency ω_n acts as a scaling factor in time , as will be proved in the next sections.

Reference

When $\nu = 1$, the transfer function reverts to the classical second-order normalized form:

$$G(s) = \frac{Ke^{-\theta s}}{1 + 2\zeta\left(\frac{s}{\omega_n}\right) + \left(\frac{s}{\omega_n}\right)^2} = \frac{K\omega_n^2 e^{-\theta s}}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{3.3}$$

This backwards compatibility ensures the identification model will hold its value for traditional integer-order systems while expanding their reach to fractional orders.

3.3 Stability Analysis of the non-commensurate system

The transition from integer-order to fractional-order systems introduces significant complexity in determining the stability criteria. Classical methods like the Routh-Hurwitz criteria become highly complex for cases such as the system studied by this article [1].

3.3.1 Stability Criteria

The work done by *Hmed et al.* [1] treats the plant as a double nested loop and assesses the stability using the Nyquist Criteria. This research's starting point will be fully based on the stability conditions found in that article. In this case, the system is stable if, and only if, the following conditions apply:

- $0 < \nu \leq 1$ and $\zeta \geq 0$;
- $0 < \nu < 1$, $\zeta < 0$ and $\frac{-2^{\nu+1}\zeta}{\cos\left(\frac{\nu\pi}{2}\right)}\left(-\zeta \tan\left(\frac{\nu\pi}{2}\right)\right)^{\nu} < 1$;
- $1 < \nu < 2$, $\zeta > 0$ and $(2\zeta\omega_n)^2\omega_u^{2\nu} + \omega_u^{2\nu+2} - \omega_n^{2\nu+2} > 0$ with $\omega_u = 2\zeta\omega_n \tan\left(\frac{(2-\nu)\pi}{2}\right)$,

Check equations in article

which result in the Stability Diagram presented in Figure 3.1.

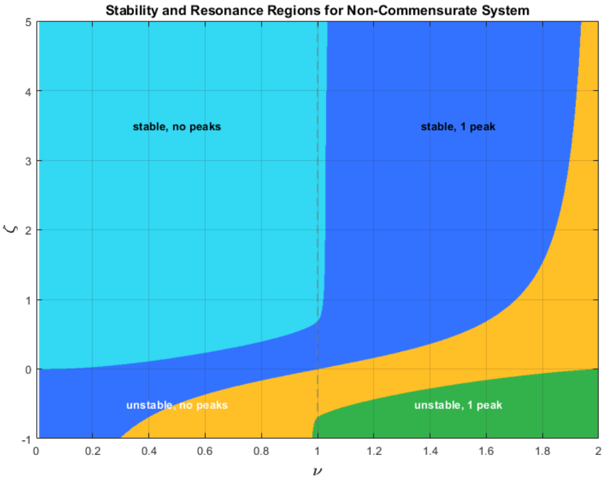

Figure 3.1: Stability Chart of the Fraction Order Transfer Function

HERE

comment the image

3.4 Time-Scaling Property of the Natural Frequency

The article [2] mentions the effect of varying ω_n is just a scale factor in time. Therefore, a fundamental step of this research is to verify if the same property applies to the non-commensurate FOTF, since it will greatly facilitate the future identification methods by reducing the parameter space by one dimension. In the next steps, an analytical approach proves this thesis.

Consider the transfer function (neglecting gain and delay):

add references

$$G(s) = \frac{1}{1 + 2\zeta \left(\frac{s}{\omega_n}\right)^{\nu} + \left(\frac{s}{\omega_n}\right)^{\nu+1}} \tag{3.4}$$

Define the normalized variable:

$$p = \frac{s}{\omega_n} \tag{3.5}$$

Then,

$$G(s) = \frac{1}{1 + 2\zeta p^{\nu} + p^{\nu+1}} = G_n(p) \tag{3.6}$$

which can be written as:

$$G(s) = G_n\left(\frac{s}{\omega_n}\right) \tag{3.7}$$

Taking the inverse Laplace transform:

check this step

$$g(t) = \mathcal{L}^{-1}\{G(s)\} \tag{3.8}$$

and using the time-scaling property:

$$g(t) = \omega_n \, g_n(\omega_n t) \tag{3.9}$$

The step response is obtained as:

$$y(t) = \int_0^t g(\tau) \, d\tau \tag{3.10}$$

Substituting $g(t)$:

$$y(t) = \int_0^t \omega_n \, g_n(\omega_n \tau) \, d\tau \tag{3.11}$$

Applying the change of variable:

$$u = \omega_n \tau, \quad d\tau = \frac{du}{\omega_n} \tag{3.12}$$

yields:

$$y(t) = \int_0^{\omega_n t} g_n(u) \, du \tag{3.13}$$

Finally,

$$y(t) = y_n(\omega_n t) \tag{3.14}$$

This formal proof shows that the natural frequency ω_n acts as a time-scaling factor, compressing or expanding the system response along the time axis without altering its geometrical proportions. Visual results of this property are displayed in Figure 3.2, generated by the file `varyingWn.m`.

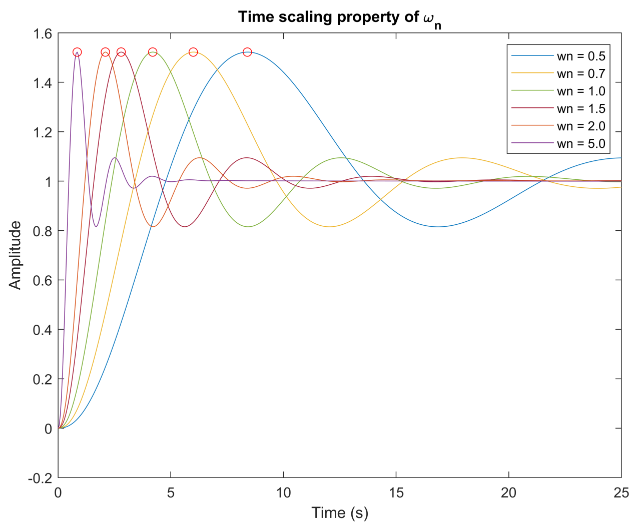

Figure 3.2: The effect of varying ω_n on different step-response curves.

As observed, while the absolute time-to-peak t_p and settling times vary with ω_n, the maximum overshoot M_p and the relative shape of the curve remain identical. The invariance justifies the adoption of a normalized database ($\omega_n = 1$) for the identification of the fractional parameters in the subsequent chapters.

3.5 Problem Statement and Goals

Despite the analytical progress made in characterizing the stability and resonance of the fractional order system, as seen in the work of *Hmed et al.* [1] and *Valério et al.* [2,3], there is still a challenge to be tackled: the identification of such systems from experimental or simulated data.

While identification for integer-order systems is a mature field with well-established methods, the nature of fractional systems require a different approach. This dissertation will aim to address this by developing an empirical Identification Model that uses extracted parameters from the unit step time response of said systems.

3.5.1 Formulation of the Problem

Let's consider that a step with amplitude L is applied to the input of the system described by Equation (3.2). From its output, $y(t)$, we know that:

- The system's gain K can be found as $\frac{1}{L}\lim_{x\to\infty} y(t)$ [2];
- The delay time θ can be found by computing the first time instant that verifies $y(t) \neq 0$. Although some issues may arise when analyzing data corrupted by noise, it doesn't pose a challenge in an ideal environment [2];
- The value of the natural frequency, ω_n, proved to be just a scale factor in time, as demonstrated in Section 3.4, therefore we can assume $\omega_n = 1$ for the sake of simplifying the system's equation.

The problem is then easily reduced to Equation (3.15):

$$G(s) = \frac{1}{1 + 2\zeta s^{\nu} + s^{\nu+1}} \tag{3.15}$$

Consequently, the first step will be to identify the fractional order, ν and the pseudo-damping coefficient, ζ, from the shape of the curve and then the natural frequency ω_n from the time values of the output.

decide on a name to call the function throughout the document

This identification strategy follows the same rationale established by previous studies [2, 4, 5], which successfully demonstrated that the parameters of given fractional-order plants can be accurately estimated by mapping geometric regularities in their reaction curves. While those works focused on Fractional First-Order and Second-Species Commensurate Order systems, this thesis extends that methodology to the non-commensurate model described in eq. (3.2) and aims to validate whether the correlation between time domain features (such as $t_{0,5}$ and M_p) and system parameters remains robust enough for this more complex fractional-order structure.

4

Numerical Methodology

Contents

The objective of this chapter is to describe the computational framework developed to generate a comprehensive database of unit step responses for the non-commensurate second species system. Given the analytical complexity of fractional calculus, this research adopts a numerical approach to bridge the gap between the theoretical transfer function and the time-domain features required for system identification.

4.1 Numerical Computation of the Step Response

The unit step response of a given system can be theoretically defined as the time-domain convolution between the input and the system's impulse response; the latter is expressed as a generalized Mittag-Leffler Function. For second species systems, a good idea is generally to obtain two first-species transfer functions by partial fraction expansion [2]. However, since the system being studied is non-commensurate, this is not possible. In this case, the analytical step response of the FOTF is expressed in eq. (4.1), as well as the general Mittag-Leffler function in (4.2).

decide if we keep K and wn in 4.1

$$y(t) = K \sum_{k=0}^{\infty} \frac{(-\omega_n^{\nu+1})^k}{k!} t^{(\nu+1)k} E^{(k)}_{\nu,(\nu+1)k+1}\left(-2\zeta\omega_n t^{\nu}\right) \tag{4.1}$$

$$E_{\alpha,\beta}(z) = \sum_{j=0}^{\infty} \frac{z^j}{\Gamma(\alpha j + \beta)} \tag{4.2}$$

However, the numerical evaluation of eq. (4.1) is computationally expensive due to the slow convergence of the power series E [2]. This burden makes the analytical approach impractical for generating large-scale step-response databases, as will be needed for the identification procedure.

4.1.1 Frequency Domain Analysis

To bypass the analytical complexity of the Mittag-Leffler Series, the system response is evaluated in the frequency domain. For a stable FOTF, the Laplace transform $Y(s)$, the output, is mapped onto the Fourier domain by the variable substitution $s \to j\omega$, provided that the system's poles lie in the left half of the complex s-plane.

The output of the system in the Laplace domain is defined as the product of the transfer function $G(s)$ and the unit step input $U(s) = \frac{1}{s}$. By applying the frequency mapping, we obtain the complex frequency response of the output, $Y(j\omega)$:

$$Y(j\omega) = \frac{1}{j\omega}\left[\frac{1}{1 + 2\zeta s^{\nu} + sagora^{\nu+1}}\right] \tag{4.3}$$

4.1.2 Numerical Inverse Fourier Transform

add references

The time-domain response $y(t)$ is reconstructed from $Y(j\omega)$ using the inverse Fourier integral. For causal systems, the relationship is given by:

$$y(t) = \mathcal{F}^{-1}\{Y(j\omega)\} = \frac{1}{2\pi} \int Y(j\omega) \cdot e^{j\omega t}\, d\omega \tag{4.4}$$

check equation. lecture notes 38.14

maybe add values

In this work, eq. (4.4) is solved numerically using the trapezoidal rule over a finite frequency range $[0, \omega_{max}]$ and $\Delta\omega = XX$, as shown in Equation (4.5).

$$y(t) \approx \frac{1}{\pi} \left[\frac{\Delta\omega}{2} \sum_{k=1}^{N} \left(f(\omega_k, t) + f(\omega_{k+1}, t) \right) \right] + \frac{1}{2} \cdot \Re\{G(0)\} \tag{4.5}$$

Where

$$f(\omega, t) = \Re\{Y(j\omega) e^{j\omega t}\} \tag{4.6}$$

explain that a database was created

This method, implemented in the custom script `invFourierTrapz.m`, allows for the efficient generation of thousands of reaction curves with high precision, regardless of the non-commensurate nature of the exponents ν and $\nu + 1$.

4.2 Characterization of the Step Response Curve

The identification of the system's parameters (ν, ζ) relies on the premise that the step response's curve profile contains unique information about the system's dynamics. Since, as stated before, the database will be generated using $\omega_n = 1$, the characterization focuses on features that describe the shape and relative timing of the curve.

4.2.1 Morphological Features of the Curve

REFERENCES

It is possible to observe, in some step responses of eq. (3.15), an overshoot and oscillations. Unlike integer-order second-order systems, where an overshoot only occurs for $\zeta < 1$, fractional-order systems can exhibit oscillations even for higher damping ratios, depending on the value of ν. This non-intuitive behavior reinforces the need for a comprehensive morphological database.

To translate the curve's geometry into a set of identification parameters, two primary types of features are extracted from each simulated curve:

1. Maximum Overshoot (M_p): Represents the peak value of the response time relative to the Steady-state (SS) value.

$$M_p = y_{peak} - y_\infty \tag{4.7}$$

2. Fractional Rise Times ($t_{x\%}$): The time instats required for the response to reach a specific percentage x of the SS value y_∞. In this work the main points selected were the following:

    - $t_{0,2}$ and $t_{0,8}$: times to reach 20% and 80% of the SS value. Used to characterize the slope of the initial rise and extract the dimensionless time ratio $\tau = \frac{t_{0,8}}{t_{0,2}}$;
    - $t_{0,5}$ (Midpoint): The time to reach 50% of the final value.

maybe isolate tau on its own point?

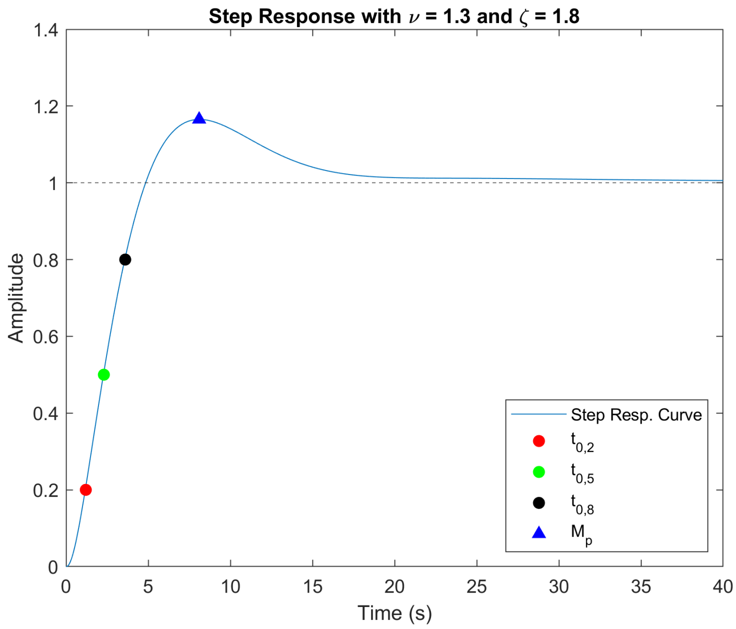

Figure 4.1: Extracted points from an example curve

Figure 4.1 displays the Unit Step Response Curve of the FOTF when $\nu = 1,3$ and $\zeta = 1,8$ and the respective interest points mentioned before.

5

This is the Fifth Chapter

6

Conclusion

7

Testing stuff

lorem ipsum bla bla bla

i did this and that and i cant forget to put a todo here and now i will see if this works.

do this and that

RF: THIS is an inline todo

Referencing:

I can reference chapter 6 or Chapter 6 or 6.

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In this other case, another co-author is making a note about the citation for missing some bibliographic record [**?**, **?**, **?**].

Pete

You should cite also Pellentesque:2014

Please note that the **FIRST LINE** of a paragraph after a **HEADING** (of Chapter or Section) **SHOULD NOT be indented**, as this is a typographical rule for this type of document.

Another rule is that you **SHOULD NEVER force LINE BREAKS** between paragraphs. You just need to leave a 'blank' line to separate paragraphs so that the document processor "understands" that the previous paragraph (independently of the number of "sentences" in it) was ended, and a new paragraphs is to be started. Note also that each paragrapgh is identified by a single LINE NUMBER in the editor pane.

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RC

use of ACRONYM

For example, the initial diameter (D_0) of the tube corresponds to a surface area (A_s) of $122\mu m^2$.

RC

use of SYMBOL defined in "Glossary"

Please notice the use of automatic referencig to objects such as Figures, Tables, equations, Algorithms, sections of a document, etc. by using the command `\Cref{ref}` as in this case pointing to **??**.

RC

auto naming of the float object, Figure

You can use in-paragraph lists with this construct for: (a) first case; (b) second case; and (c) third case, making the text organized and fluid.

Notice that mathematics makes extensive use of Formulas which are particularly well rendered in documents produced with LaTeX.

RC

example of use of Glossaries

return value type | function name | list of formal parameters

```
int puissance (int x, int n) {

    int i, p = 1;

    for (i = 1; i <= n; i++)
      p = p * x;

    return p;
}
```

local variables declaration

instructions

instruction `return`

Listing 7.1: A listing with a Tikz picture overlayed

Algorithm 7.1: Time Control Strategy

begin
- $nextBitrate \longleftarrow nextDownloadLevel$
- $nextBitrate \longleftarrow GetNextBitrate()$
- $cpuLoad \longleftarrow GetCpuLoad()$
- $bitrateDelta \longleftarrow getBitrateDelta(currentBitrate, nextBitrate)$
- **if** $bitrateDelta > maxThreshold$ **then**
  - $SetBitrate(nextBitrate)$
- **if** $minThreshold < bitrateDelta < maxThreshold$ **and** $numAttemps < 2$ **then**
  - $numAttemps \longleftarrow numAttemps + 1$
- **else if** $minThreshold < bitrateDelta < maxThreshold$ **and** $numAttemps = 2$ **then**
  - $numAttemps \longleftarrow 0$
- **else**
  - $SetBitrate(nextBitrate)$
- **if** $0 < bitrateDelta < minThreshold$ **and** $numAttemps < 3$ **then**
  - $numAttemps \longleftarrow numAttemps + 1$
- **else if** $0 < bitrateDelta < minThreshold$ **and** $numAttemps = 3$ **then**
  - $SetBitrate(nextBitrate)$

Table 7.1 is automatically compressed to fit text width. You can use `https://www.tablesgenerator.com` to produce these tables, and then copy the LaTeX code generated to paste in the document.

Table 7.1: Comparison between today's and target Architectures of Telcos

Today		Target	
Rigid	Each evolutionary requirement involves development of multiple components, interfaces, platforms,etc.	Flexible	It is possible to modify or add new functionalities rapidly.
Slow	Development of a new application takes months or years.	Fast	Development of a new application takes weeks instead of months or years.
Closed	Limited integration with external environments.	Open	It is simple to integrate internal, applications with external entities.
Complex	Heterogeneous technologies, obsolescence, lack,of standards, high redundancy.	Standardised	Use of homogeneous architectural models.
Expensive	High Capex (for new service development) and,high,Opex (to ensure running of IT).	Cost-Effective	Capex and Opex are optimised.

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[5] J. J. Gude and P. García Bringas, "Proposal of a General Identification Method for Fractional-Order Processes Based on the Process Reaction Curve," *Fractal and Fractional*, vol. 6, no. 9, p. 526, Sep. 2022.

List of Symbols

A_s	Surface Area	μm^2
D_0	Initial Diameter	μm
M_p	Maximum Overshoot	
ν	Real-valued Laplace variable exponent	
ω_n	Natural Frequency	rad/s
ζ	Pseudo-damping Coefficient	
s	Laplace Variable	s^{-1}
t_p	Time to the Maximum Overshoot	
$t_{0,5}$	Time to reach 50% of the steady-state yield	s
y_∞	Steady-state Value	

Glossary

formula

A mathematical expression 17

LaTeX

LaTeX It is a mark up language specially suited for scientific documents as it can correctly format documents with all the typographical rules 17

mathematics

Mathematics is what mathematicians do 17

A

Code of Project

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Listing A.1: Example of a XML file.

```xml
<?xml version="1.0" encoding="UTF−8"?>
<StreamInfo version="2.0">
    <Clip duration="PT01M0.00S">
        <BaseURL>videos/</BaseURL>
        <Description>svc_1</Description>
        <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="401.90"
            width="176" height="144" id="L0">
            <BaseURL>svc_1/</BaseURL>
            <SegmentInfo from="0" to="11" duration="PT5.00S">
                <BaseURL>svc_1−L0−</BaseURL>
            </SegmentInfo>
        </Representation>
        <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="1322.60"
            width="352" height="288" id="L1">
            <BaseURL>svc_1/</BaseURL>
            <SegmentInfo from="0" to="11" duration="PT5.00S">
```

```
            <BaseURL>svc_1-L1-</BaseURL>
          </SegmentInfo>
        </Representation>
      </Clip>
    </StreamInfo>
```

Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed cursus cursus velit. Sed a massa. Duis dignissim euismod quam.

Listing A.2: Assembler Main Code.

```
; *********************************************************************
; * Constantes
; *********************************************************************

ON     EQU 1 ; contagem ligada
OFF    EQU 0 ; contagem desligada
INPUT EQU 8000H ; endereco do porto de entrada
          ;(bit 0 = RTC; bit 1 = botao)
OUTPUT  EQU 8000H ; endereco do porto de saida.


; *********************************************************************
; * Stack
; *********************************************************************

PLACE   1000H
pilha:    TABLE 100H  ; espaco reservado para a pilha
fim_pilha:

; *********************************************************************

PLACE   2000H

; Tabela de vectores de interrupcao

tab:    WORD  rot0

; *********************************************************************
; * Programa Principal
; *********************************************************************

PLACE   0

inicio:
  MOV BTE, tab    ; incializa BTE
  MOV R9, INPUT   ; endereco do porto de entrada
  MOV R10, OUTPUT   ; endereco do porto de saida
  MOV SP, fim_pilha
  MOV R5, 1     ; inicializa estado do processo P1
  MOV R6, 1     ; inicializa estado do processo P2
  MOV R4, OFF     ; inicializa controle de RTC
  MOV R8, 0     ; inicializa contador
  MOV R7, OFF     ; inicialmente nao permite contagem
  EI0         ; permite interrupcoes tipo 0
  EI          ; activa interrupcoes

ciclo:
  CALL  P1      ; invoca processo P1
  CALL  P2      ; invoca processo P2
  JMP   ciclo   ; repete ciclo

; *********************************************************************
;* ROTINAS
; *********************************************************************

P1:
  CMP R5, 1   ; se estado = 1
  JZ  P1_1
  CMP R5, 2   ; se estado = 2
```

```
  JZ   P1_2
sai_P1:
  RET        ; sai do processo.


P1_1:
  MOVB   R0, [R9]  ; ler porto de entrada
  BIT R0, 1
  JZ   sai_P1     ; se botao nao carregado, sai do processo
  MOV R7, ON      ; permite contagem do display
  MOV R5, 2    ; passa ao estado 2 do P1
  JMP sai_P1

P1_2:
  MOVB   R0, [R9]  ; ler porto de entrada
  BIT R0, 1
  JNZ sai_P1     ; se botao continua carregado, sai do processo
  MOV R7, OFF    ; caso contrario, desliga contagem do display
  MOV R5, 1    ; passa ao estado 1 do P1
  JMP sai_P1
```

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus eget nisl ut elit porta ullamcorper. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos.

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Listing A.3: HTML with CSS Code

```
<!DOCTYPE html>
<html>
  <head>
    <title>Listings Style Test</title>
    <meta charset="UTF-8">
    <style>
      /* CSS Test */
      * {
        padding: 0;
        border: 0;
        margin: 0;
      }
    </style>
    <link rel="stylesheet" href="css/style.css" />
  </head>
  <header> hey </header>
  <article> this is a article </article>
  <body>
    <!-- Paragraphs are fine -->
```

```
  <div id="box">
    <p>
      Hello World
    </p>
    <p>Hello World</p>
    <p id="test">Hello World</p>
    <p></p>
  </div>
  <div>Test</div>
  <!-- HTML script is not consistent -->
  <script src="js/benchmark.js"></script>
  <script>
    function createSquare(x, y) {
      // This is a comment.
      var square = document.createElement('div');
      square.style.width = square.style.height = '50px';
      square.style.backgroundColor = 'blue';

      /*
       * This is another comment.
       */
      square.style.position = 'absolute';
      square.style.left = x + 'px';
      square.style.top = y + 'px';

      var body = document.getElementsByTagName('body')[0];
      body.appendChild(square);
    };

    // Please take a look at +=
    window.addEventListener('mousedown', function(event) {
      // German umlaut test: Berührungspunkt ermitteln
      var x = event.touches[0].pageX;
      var y = event.touches[0].pageY;
      var lookAtThis += 1;
    });
  </script>
</body>
```

```
</html>
```

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Listing A.4: HTML CSS Javascript Code

```

@media only screen and (min-width: 768px) and (max-width: 991px) {

  #main {
    width: 712px;
    padding: 100px 28px 120px;
  }

  /* .mono {
    font-size: 90%;
  } */

  .cssbtn a {
    margin-top: 10px;
    margin-bottom: 10px;
    width: 60px;
    height: 60px;
    font-size: 28px;
    line-height: 62px;
  }
```

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Listing A.5: PYTHON Code

```python
class TelgramRequestHandler(object):
    def handle(self):
        addr = self.client_address[0]         # Client IP-adress
        telgram = self.request.recv(1024)     # Recieve telgram
        print "From: %s, Received: %s" % (addr, telgram)
        return
```

B

A Large Table

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As Table B.1 shows, the data can be inserted from a file, in the case of a somehow complex structure. Notice the Table footnotes.

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Table B.1: Example table

Benchmark: ANN	**#Layers** (1)	**#Nets** (2)	#Nodes* $(3) = 8 \cdot (1) \cdot (2)$	Critical path $(4) = 4 \cdot (1)$	**Latency** **(T_{iter})** (5)
A1	**3–1501**	1	**24–12008**	**12–6004**	4
A2	501	1	4008	2004	**2–2000**
A3	10	**2–1024**	**160–81920**	40	60†
A4	10	50	4000	40	**80–1200**
Benchmark: FFT	**FFT size‡** (1)	*#Inputs* $(2) = 2^{(1)}$	*#Nodes** $(3) = 10 \cdot (1) \cdot (2)$	*Critical path* $(4) = 4 \cdot (1)$	**Latency** **(T_{iter})** (5)
F1	**1–10**	2–1024	**20–102400**	4–40	6–60†
F2	**5**	32	1600	20	**40 – 1500**
Benchmark: Random networks	**#Types** (1)	**#Nodes** (2)	*#Networks* (3)	*Critical path* (4)	**Latency** **(T_{iter})** (5)
R1	3	10–2000	500	*variable*	(4)
R2	3	50	500	*variable*	$(4) \times [1; \cdots ; 20]$

* *Excluding constant nodes.*
† *Value kept proportional to the critical path:* $(5) = (4) * 1.5$.
‡ *A size of* x *corresponds to a* 2^x *point FFT.*
Values in bold indicate the parameter being varied.

pharetra posuere sapien. Nam consectetuer. Sed aliquam, nunc eget euismod ullamcorper, lectus nunc ullamcorper orci, fermentum bibendum enim nibh eget ipsum. Donec porttitor ligula eu dolor. Maecenas vitae nulla consequat libero cursus venenatis. Nam magna enim, accumsan eu, blandit sed, blandit a, eros.

And now an example (Table B.2) of a table that extends to more than one page. Notice the repetition of the Caption (with indication that is continued) and of the Header, as well as the continuation text at the bottom.

Table B.2: Example of a very long table spreading in several pages

Time (s)	Triple chosen	Other feasible triples
0	(1, 11, 13725)	(1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0)
2745	(1, 12, 10980)	(1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0)
5490	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
8235	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
10980	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
13725	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
16470	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
19215	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
21960	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
24705	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
27450	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
30195	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
32940	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

Continued on next page...

Table B.2: (...continued from previous page)

Time (s)	Triple chosen	Other feasible triples
35685	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
38430	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
41175	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
43920	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
46665	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
49410	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
52155	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
54900	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
57645	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
60390	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
63135	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
65880	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
68625	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
71370	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
74115	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
76860	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
79605	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
82350	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
85095	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
87840	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
90585	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
93330	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
96075	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
98820	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
101565	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
104310	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
107055	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
109800	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
112545	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
115290	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
118035	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
120780	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
123525	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
126270	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
129015	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
131760	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
134505	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
137250	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
139995	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
142740	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
145485	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
148230	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
150975	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
153720	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
156465	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
159210	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
161955	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
164700	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

An example of a large Table that autofits the size to the page margins is illustrated in Table B.3. Please notice the text size that is shrunken in order fot the table to adjust to the page:

Table B.3: Sample Table.

URL	First Time Visit	Last Time Visit	URL Counts	Value	Reference
https://web.facebook.com/	1521241972	1522351859	177	56640	[facebook-2021]
http://localhost/phpmyadmin/	1518413861	1522075694	24	39312	database-management
https://mail.google.com/mail/u/	1516596003	1522352010	36	33264	Google-Gmail-2021
https://github.com/shawon100	1517215489	1522352266	37	27528	Code-Repository
https://www.youtube.com/	1517229227	1521978502	24	14792	Youtube-video-2021